

```
root@vpn.unixclinic.net # . vars
```



Note: If you run `./clean-all`, it will be equal to an `rm -rf /opt/easy-rsa/keys/`

```
root@vpn.unixclinic.net # ./clean-all
```

Next we need to build the Diffie-Hellman (DH) parameters. It took me less than a minute to generate a 1024-bit long safe prime on my server—I have snipped the output:

```
root@vpn.unixclinic.net # build-dh
```

```
Generating DH parameters, 1024 bit long safe prime, generator 2
This is going to take a long time
```

```
.....+
```

```
... [snipped.]
```

Next we will generate the root CA certificate and key:

```
root@vpn.unixclinic.net # build-ca
```

```
Generating a 1024 bit RSA private key
```

```
..+++++
```

```
..+++++
```

```
writing new private key to 'ca.key'
```

```
.....
```

You are about to be asked to enter information that will be incorporated into your certificate request.

What you are about to enter is what is called a Distinguished Name or a DN.

There are quite a few fields but you can leave some blank

For some fields there will be a default value,

If you enter '.', the field will be left blank.

```
.....
```

```
Country Name (2 letter code) [IN]:
```

```
State or Province Name (full name) [UA]:
```

```
Locality Name (eg. city) [Nainital]:
```

```
Organization Name (eg. company) [Unixclinic]:
```

```
Organizational Unit Name (eg. section) []:
```

```
Common Name (eg. your name or your server's hostname) [Unixclinic CA]:
```

```
unixclinic.net
```

```
Name []: Unixclinic CA
```

```
Email Address [casupport@unixclinic.net]:
```

Now that our CA has been set up, we need to create a certificate for our VPN server. This can be done as follows:

```
root@vpn.unixclinic.net # build-key-server vpn.unixclinic.net
```

```
Generating a 1024 bit RSA private key
```

```
.....+++++
```

```
..+++++
```

```
writing new private key to 'vpn.unixclinic.net.key'
```

```
.....
```

You are about to be asked to enter information that will be incorporated into your certificate request.

What you are about to enter is what is called a Distinguished Name or a DN.

There are quite a few fields but you can leave some blank

For some fields there will be a default value,

If you enter '.', the field will be left blank.

```
.....
```

```
Country Name (2 letter code) [IN]:
```

```
State or Province Name (full name) [UA]:
```

```
Locality Name (eg. city) [Nainital]:
```

```
Organization Name (eg. company) [Unixclinic]:
```

```
Organizational Unit Name (eg. section) []:
```

```
Common Name (eg. your name or your server's hostname) [vpn.unixclinic.net]:
```

```
Name []:
```

```
Name []: OpenVPN Server
```

```
Email Address [casupport@unixclinic.net]: vpnsupport@unixclinic.net
```

Please enter the following 'extra' attributes

to be sent with your certificate request

A challenge password []: password

An optional company name []: Unixclinic

Using configuration from /opt/easy-rsa/openssl.cnf

Check that the request matches the signature

Signature ok

The Subject's Distinguished Name is as follows

```
countryName             PRINTABLE: 'IN'
```

```
stateOrProvinceName     PRINTABLE: 'UA'
```

```
localityName            PRINTABLE: 'Nainital'
```

```
organizationName       PRINTABLE: 'Unixclinic'
```

```
commonName             PRINTABLE: 'vpn.unixclinic.net'
```

```
name                  PRINTABLE: 'OpenVPN Server'
```

```
emailAddress           1ASSTRING: 'vpnsupport@unixclinic.net'
```

Certificate is to be certified until Jun 22 04:24:13 2010 GMT (365 days)

Sign the certificate? [y/n]: y

1 out of 1 certificate requests certified, commit? [y/n]

Write out database with 1 new entries

Database Updated

Now we will have to copy the server certificate to the `/etc/openvpn/keys` directory and make sure that the private key has the most restrictive permissions:

```
root@vpn.unixclinic.net # cp -v /opt/easy-rsa/keys/vpn.unixclinic
```

```
net.crt.key /opt/easy-rsa/keys/dh1024.pem /etc/openvpn/keys/
```

```
'/opt/easy-rsa/keys/vpn.unixclinic.net.crt' -> '/etc/openvpn/keys/vpn.unixclinic.net.crt'
```

```
'/opt/easy-rsa/keys/vpn.unixclinic.net.key' -> '/etc/openvpn/keys/vpn.unixclinic.net.key'
```

```
'/opt/easy-rsa/keys/dh1024.pem' -> '/etc/openvpn/keys/dh1024.pem'
```

```
root@vpn.unixclinic.net # cp -v /opt/easy-rsa/keys/ca.crt /etc/openvpn/keys/
```